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```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.metrics import confusion_matrix, classification_report
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, BatchNormalization

iris = datasets.load_iris()
X = iris.data
y = iris.target

scaler = StandardScaler()
X = scaler.fit_transform(X)

encoder = LabelEncoder()
y = encoder.fit_transform(y)
y = keras.utils.to_categorical(y, 3)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = Sequential([
    Dense(8, activation='relu', input_shape=(4,)),
    BatchNormalization(),
    Dense(8, activation='relu'),
    Dropout(0.2),
    Dense(3, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

history = model.fit(X_train, y_train, validation_data=(X_test, y_test), epochs=100, batch_size=8, verbose=1)

loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print(f"Test Accuracy: {accuracy:.4f}")

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()

y_pred = np.argmax(model.predict(X_test), axis=1)
y_true = np.argmax(y_test, axis=1)
cm = confusion_matrix(y_true, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

print("Classification Report:\n", classification_report(y_true, y_pred))
```



```
ython3.11/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` arg
__(activity_regularizer=activity_regularizer, **kwargs)
----- 4s 44ms/step - accuracy: 0.3224 - loss: 1.3262 - val_accuracy: 0.0333 - val_loss: 1.2824

----- 1s 20ms/step - accuracy: 0.4729 - loss: 0.9999 - val_accuracy: 0.1333 - val_loss: 1.2209

----- 0s 16ms/step - accuracy: 0.4049 - loss: 1.1314 - val_accuracy: 0.2000 - val_loss: 1.1542

----- 0s 11ms/step - accuracy: 0.4310 - loss: 1.0026 - val_accuracy: 0.4333 - val_loss: 1.0842

----- 0s 16ms/step - accuracy: 0.5861 - loss: 0.9779 - val_accuracy: 0.5333 - val_loss: 1.0120

----- 1s 40ms/step - accuracy: 0.6094 - loss: 0.8852 - val_accuracy: 0.6000 - val_loss: 0.9386

----- 0s 11ms/step - accuracy: 0.6414 - loss: 0.7949 - val_accuracy: 0.6667 - val_loss: 0.8660

----- 0s 16ms/step - accuracy: 0.6479 - loss: 0.7833 - val_accuracy: 0.6667 - val_loss: 0.8029

----- 0s 8ms/step - accuracy: 0.7234 - loss: 0.7253 - val_accuracy: 0.7333 - val_loss: 0.7420

----- 0s 10ms/step - accuracy: 0.7016 - loss: 0.6934 - val_accuracy: 0.8667 - val_loss: 0.6842

----- 0s 17ms/step - accuracy: 0.7392 - loss: 0.7395 - val_accuracy: 0.9000 - val_loss: 0.6326

----- 0s 10ms/step - accuracy: 0.7558 - loss: 0.6353 - val_accuracy: 0.9000 - val_loss: 0.5859

----- 0s 11ms/step - accuracy: 0.7705 - loss: 0.6065 - val_accuracy: 0.9000 - val_loss: 0.5452

----- 0s 23ms/step - accuracy: 0.7812 - loss: 0.5849 - val_accuracy: 0.9000 - val_loss: 0.5081

----- 1s 22ms/step - accuracy: 0.7675 - loss: 0.5803 - val_accuracy: 0.9333 - val_loss: 0.4778

----- 0s 14ms/step - accuracy: 0.8577 - loss: 0.4384 - val_accuracy: 0.9333 - val_loss: 0.4454

----- 0s 20ms/step - accuracy: 0.7839 - loss: 0.4839 - val_accuracy: 0.9333 - val_loss: 0.4147

----- 1s 32ms/step - accuracy: 0.7462 - loss: 0.6337 - val_accuracy: 0.9000 - val_loss: 0.3911

----- 0s 23ms/step - accuracy: 0.7734 - loss: 0.5867 - val_accuracy: 0.9333 - val_loss: 0.3684

----- 0s 26ms/step - accuracy: 0.7873 - loss: 0.6055 - val_accuracy: 0.9333 - val_loss: 0.3533

----- 0s 26ms/step - accuracy: 0.8115 - loss: 0.5221 - val_accuracy: 0.9333 - val_loss: 0.3354

----- 1s 26ms/step - accuracy: 0.7661 - loss: 0.5311 - val_accuracy: 0.9667 - val_loss: 0.3160

----- 0s 13ms/step - accuracy: 0.7462 - loss: 0.5718 - val_accuracy: 0.9667 - val_loss: 0.3019

----- 0s 20ms/step - accuracy: 0.8388 - loss: 0.4448 - val_accuracy: 0.9667 - val_loss: 0.2863

----- 1s 33ms/step - accuracy: 0.8913 - loss: 0.3850 - val_accuracy: 0.9667 - val_loss: 0.2714

----- 1s 29ms/step - accuracy: 0.7860 - loss: 0.4690 - val_accuracy: 0.9667 - val_loss: 0.2622

----- 0s 14ms/step - accuracy: 0.7870 - loss: 0.4429 - val_accuracy: 0.9667 - val_loss: 0.2551

----- 0s 13ms/step - accuracy: 0.8303 - loss: 0.4556 - val_accuracy: 0.9667 - val_loss: 0.2470

----- 0s 25ms/step - accuracy: 0.8614 - loss: 0.3655 - val_accuracy: 0.9667 - val_loss: 0.2376

----- 0s 11ms/step - accuracy: 0.8335 - loss: 0.4556 - val_accuracy: 0.9667 - val_loss: 0.2300

----- 0s 10ms/step - accuracy: 0.9138 - loss: 0.3657 - val_accuracy: 1.0000 - val_loss: 0.2176

----- 0s 5ms/step - accuracy: 0.8783 - loss: 0.4080 - val_accuracy: 0.9667 - val_loss: 0.2088

----- 0s 5ms/step - accuracy: 0.8604 - loss: 0.3634 - val_accuracy: 1.0000 - val_loss: 0.1971

----- 0s 8ms/step - accuracy: 0.8717 - loss: 0.3991 - val_accuracy: 1.0000 - val_loss: 0.1889

----- 0s 5ms/step - accuracy: 0.8811 - loss: 0.3145 - val_accuracy: 1.0000 - val_loss: 0.1833

----- 0s 4ms/step - accuracy: 0.8732 - loss: 0.3414 - val_accuracy: 1.0000 - val_loss: 0.1782

----- 0s 5ms/step - accuracy: 0.8804 - loss: 0.3746 - val_accuracy: 1.0000 - val_loss: 0.1736

----- 0s 8ms/step - accuracy: 0.8680 - loss: 0.3466 - val_accuracy: 1.0000 - val_loss: 0.1687

----- 0s 4ms/step - accuracy: 0.8049 - loss: 0.4567 - val_accuracy: 1.0000 - val_loss: 0.1680

----- 0s 5ms/step - accuracy: 0.8240 - loss: 0.4468 - val_accuracy: 1.0000 - val_loss: 0.1642

----- 0s 5ms/step - accuracy: 0.9054 - loss: 0.2964 - val_accuracy: 1.0000 - val_loss: 0.1561

----- 0s 5ms/step - accuracy: 0.9142 - loss: 0.2830 - val_accuracy: 1.0000 - val_loss: 0.1528

----- 0s 5ms/step - accuracy: 0.7800 - loss: 0.4143 - val_accuracy: 1.0000 - val_loss: 0.1520

----- 0s 5ms/step - accuracy: 0.8322 - loss: 0.4762 - val_accuracy: 1.0000 - val_loss: 0.1515
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0s 5ms/step - accuracy: 0.8693 - loss: 0.4458 - val_accuracy: 1.0000 - val_loss: 0.1494

0s 5ms/step - accuracy: 0.8666 - loss: 0.3614 - val_accuracy: 1.0000 - val_loss: 0.1502

0s 4ms/step - accuracy: 0.8413 - loss: 0.3891 - val_accuracy: 1.0000 - val_loss: 0.1451

0s 5ms/step - accuracy: 0.8314 - loss: 0.4446 - val_accuracy: 1.0000 - val_loss: 0.1410

0s 5ms/step - accuracy: 0.9086 - loss: 0.4536 - val_accuracy: 1.0000 - val_loss: 0.1367

0s 5ms/step - accuracy: 0.9242 - loss: 0.2465 - val_accuracy: 1.0000 - val_loss: 0.1368

0s 6ms/step - accuracy: 0.9396 - loss: 0.2497 - val_accuracy: 1.0000 - val_loss: 0.1295

0s 5ms/step - accuracy: 0.9136 - loss: 0.2425 - val_accuracy: 1.0000 - val_loss: 0.1218

0s 4ms/step - accuracy: 0.9216 - loss: 0.2754 - val_accuracy: 1.0000 - val_loss: 0.1219

0s 4ms/step - accuracy: 0.9084 - loss: 0.2454 - val_accuracy: 1.0000 - val_loss: 0.1140

0s 5ms/step - accuracy: 0.9560 - loss: 0.2115 - val_accuracy: 1.0000 - val_loss: 0.1120

0s 5ms/step - accuracy: 0.8913 - loss: 0.2957 - val_accuracy: 1.0000 - val_loss: 0.1122

0s 5ms/step - accuracy: 0.9084 - loss: 0.2583 - val_accuracy: 1.0000 - val_loss: 0.1099

0s 5ms/step - accuracy: 0.8958 - loss: 0.2943 - val_accuracy: 1.0000 - val_loss: 0.1098

0s 4ms/step - accuracy: 0.9562 - loss: 0.2073 - val_accuracy: 1.0000 - val_loss: 0.1076

0s 5ms/step - accuracy: 0.9497 - loss: 0.2294 - val_accuracy: 1.0000 - val_loss: 0.1057

0s 5ms/step - accuracy: 0.9484 - loss: 0.2081 - val_accuracy: 1.0000 - val_loss: 0.0995

0s 5ms/step - accuracy: 0.8722 - loss: 0.3266 - val_accuracy: 1.0000 - val_loss: 0.0952

0s 5ms/step - accuracy: 0.9546 - loss: 0.2174 - val_accuracy: 1.0000 - val_loss: 0.0952

0s 5ms/step - accuracy: 0.9440 - loss: 0.2049 - val_accuracy: 1.0000 - val_loss: 0.0929

0s 5ms/step - accuracy: 0.9393 - loss: 0.2037 - val_accuracy: 1.0000 - val_loss: 0.0899

0s 5ms/step - accuracy: 0.8830 - loss: 0.2856 - val_accuracy: 1.0000 - val_loss: 0.0889

0s 5ms/step - accuracy: 0.9339 - loss: 0.2178 - val_accuracy: 1.0000 - val_loss: 0.0894

0s 5ms/step - accuracy: 0.9343 - loss: 0.2579 - val_accuracy: 1.0000 - val_loss: 0.0876

0s 5ms/step - accuracy: 0.9204 - loss: 0.2014 - val_accuracy: 1.0000 - val_loss: 0.0900

0s 5ms/step - accuracy: 0.9064 - loss: 0.2303 - val_accuracy: 1.0000 - val_loss: 0.0960

0s 5ms/step - accuracy: 0.9002 - loss: 0.2441 - val_accuracy: 1.0000 - val_loss: 0.0956

0s 5ms/step - accuracy: 0.9571 - loss: 0.1878 - val_accuracy: 1.0000 - val_loss: 0.0993

0s 5ms/step - accuracy: 0.9379 - loss: 0.1602 - val_accuracy: 0.9667 - val_loss: 0.1028

0s 5ms/step - accuracy: 0.9728 - loss: 0.1462 - val_accuracy: 0.9667 - val_loss: 0.0957

0s 5ms/step - accuracy: 0.9549 - loss: 0.2179 - val_accuracy: 0.9667 - val_loss: 0.0965

0s 5ms/step - accuracy: 0.8931 - loss: 0.3177 - val_accuracy: 0.9333 - val_loss: 0.1036

0s 5ms/step - accuracy: 0.8599 - loss: 0.3395 - val_accuracy: 0.9667 - val_loss: 0.0996

0s 5ms/step - accuracy: 0.9900 - loss: 0.1321 - val_accuracy: 0.9333 - val_loss: 0.1100

0s 5ms/step - accuracy: 0.9306 - loss: 0.1839 - val_accuracy: 0.9333 - val_loss: 0.1054

0s 4ms/step - accuracy: 0.9407 - loss: 0.2220 - val_accuracy: 0.9333 - val_loss: 0.1067

0s 4ms/step - accuracy: 0.9304 - loss: 0.3737 - val_accuracy: 0.9333 - val_loss: 0.1050

0s 8ms/step - accuracy: 0.8706 - loss: 0.4655 - val_accuracy: 0.9667 - val_loss: 0.1024

0s 9ms/step - accuracy: 0.9664 - loss: 0.1636 - val_accuracy: 0.9667 - val_loss: 0.1018

0s 9ms/step - accuracy: 0.9175 - loss: 0.1895 - val_accuracy: 0.9333 - val_loss: 0.0959

0s 10ms/step - accuracy: 0.9269 - loss: 0.1662 - val_accuracy: 0.9333 - val_loss: 0.0959

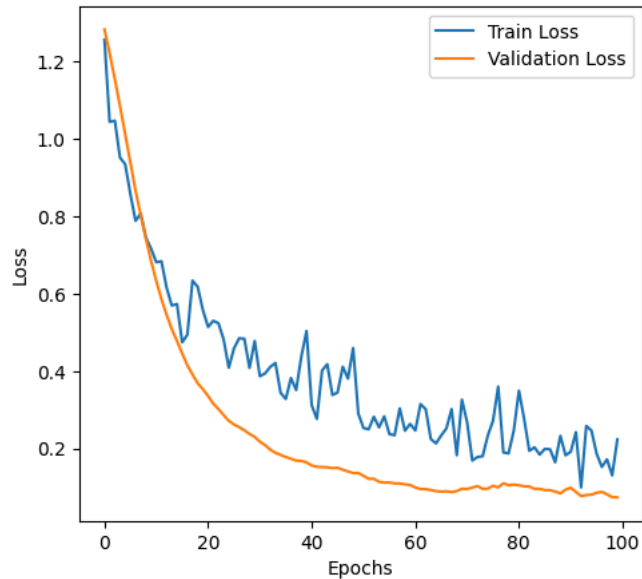
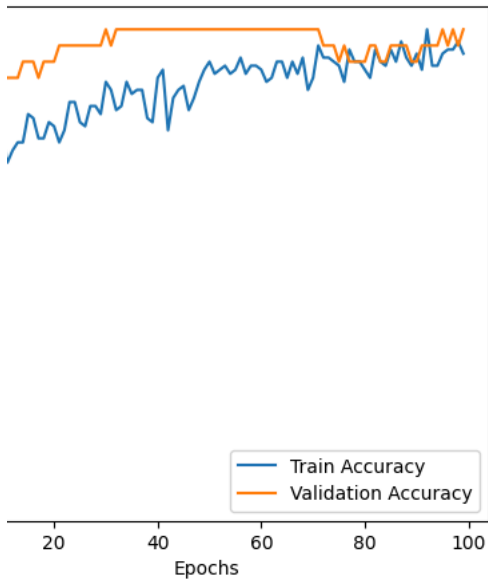
0s 9ms/step - accuracy: 0.9467 - loss: 0.2731 - val_accuracy: 0.9667 - val_loss: 0.0928

0s 10ms/step - accuracy: 0.9266 - loss: 0.2257 - val_accuracy: 0.9667 - val_loss: 0.0923

0s 5ms/step - accuracy: 0.9892 - loss: 0.1350 - val_accuracy: 0.9667 - val_loss: 0.0889

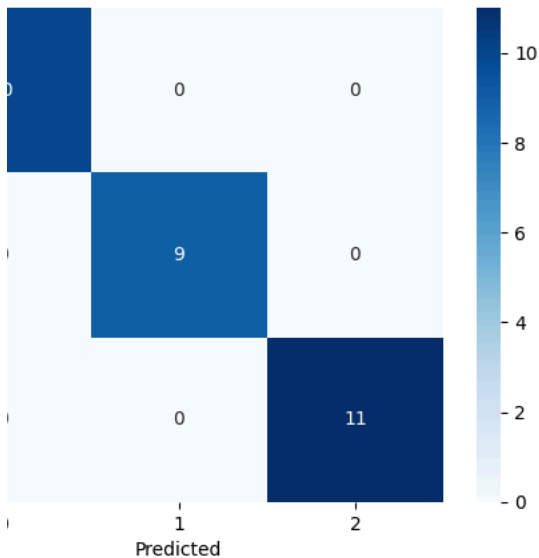
0s 4ms/step - accuracy: 0.9757 - loss: 0.1281 - val_accuracy: 0.9667 - val_loss: 0.0841

```
0s 5ms/step - accuracy: 0.9359 - loss: 0.1481 - val_accuracy: 0.9333 - val_loss: 0.0938
0s 5ms/step - accuracy: 0.9812 - loss: 0.1081 - val_accuracy: 0.9333 - val_loss: 0.0988
0s 4ms/step - accuracy: 0.9052 - loss: 0.2739 - val_accuracy: 0.9667 - val_loss: 0.0878
0s 5ms/step - accuracy: 1.0000 - loss: 0.0984 - val_accuracy: 0.9667 - val_loss: 0.0772
0s 5ms/step - accuracy: 0.9403 - loss: 0.2318 - val_accuracy: 0.9667 - val_loss: 0.0798
0s 5ms/step - accuracy: 0.8957 - loss: 0.3806 - val_accuracy: 0.9667 - val_loss: 0.0809
0s 5ms/step - accuracy: 0.9694 - loss: 0.1626 - val_accuracy: 1.0000 - val_loss: 0.0857
0s 5ms/step - accuracy: 0.9528 - loss: 0.1552 - val_accuracy: 0.9667 - val_loss: 0.0881
0s 5ms/step - accuracy: 0.9781 - loss: 0.1119 - val_accuracy: 1.0000 - val_loss: 0.0817
0s 8ms/step - accuracy: 0.9565 - loss: 0.1650 - val_accuracy: 0.9667 - val_loss: 0.0744
0s 11ms/step - accuracy: 0.9703 - loss: 0.2254 - val_accuracy: 1.0000 - val_loss: 0.0739
.0000
```



0s 92ms/step

Confusion Matrix



report:			
recision	recall	f1-score	support
1.00	1.00	1.00	10
1.00	1.00	1.00	9
1.00	1.00	1.00	11
		1.00	30
1.00	1.00	1.00	30
1.00	1.00	1.00	30

